



# Teacher Implementation Guide

### Implementation Guide

Audience: Teachers | Grades 8-12

#### Purpose

- This guide helps teachers introduce cybersecurity and digital ethics in a practical, student-safe way. It can be used as a short mini-unit, advisory sequence, or starting point for Computer Science Foundations and cybersecurity pathway instruction.

#### Standards summary

- This resource may support CSF 3.1, CSF 13.1, CSF 14.1, CSF 14.2, CSF 14.3.
- Detailed alignment depends on local district pacing, approved course placement, and teacher directions.

## Implementation Purpose

This guide helps teachers introduce cybersecurity and digital ethics in a practical, student-safe way. It can be used as a short mini-unit, advisory sequence, or starting point for Computer Science Foundations and cybersecurity pathway instruction.

## Implementation Goals

- Teach students to recognize common digital risks without teaching unsafe tactics.
- Build habits around privacy, data protection, account security, and safe response.
- Connect cybersecurity to career readiness and real-world technology decisions.
- Create consistent language for ethical AI and cybersecurity discussions.
- Help students understand that cybersecurity is about protection, responsibility, and trust.



## Recommended Sequence

| When    | Activity                             | Teacher Focus                          |
|---------|--------------------------------------|----------------------------------------|
| Day 1   | Lesson: Spot the Phish               | Recognition, red flags, safe response  |
| Day 2   | Password Strength Activity           | Account security, MFA, passphrases     |
| Day 3   | Digital Footprint Reflection         | Privacy, data sharing, online identity |
| Day 4   | Cybersecurity Careers One-Pager      | Career exploration and skills          |
| Ongoing | AI + Cybersecurity Discussion Prompt | Ethics, AI risks, defensive use        |

## Classroom Norms to Establish

- All cybersecurity work must be defensive, ethical, and teacher-approved.
- Students should not use real accounts, real passwords, real targets, or live suspicious links.
- Students should not attempt to bypass security controls.
- Students should not share private information with AI tools.
- Students should report concerns through teacher-approved or school-approved channels.
- Students should understand the difference between learning security concepts and misusing security knowledge.

## Suggested Teacher Language

In this class, cybersecurity means learning how to protect people, systems, and information. We will study risks, safe habits, privacy, and ethical decision-making. We will not attack systems, target real people, or practice unsafe behavior.

## Administrator-Facing Value

- Supports safe cybersecurity awareness without offensive hacking activities.
- Connects Computer Science Foundations to cybersecurity pathway readiness.
- Builds student understanding of privacy, data security, and ethical technology use.
- Provides practical classroom routines administrators can understand and support.
- Creates a bridge from AI literacy to cybersecurity and digital ethics.

## Family Communication Option

Families can be told that students are learning how to protect accounts, recognize suspicious messages, think carefully about digital footprints, and use AI and cybersecurity knowledge responsibly. The goal is not to teach students how to break systems. The goal is to help students make safer, more ethical technology decisions.



# Detailed Tennessee Standards Connection

## Tennessee Standards Connection

This implementation guide helps teachers connect cybersecurity awareness to data security, privacy, ethical technology use, career awareness, and classroom-safe routines.

Standards source: Tennessee Department of Education, Computer Science Foundations (C10H11), May 2023. Confirm final alignment against local district pacing, approved course placement, and teacher directions.

This resource may support the following Tennessee standards when used as part of cybersecurity, digital ethics, computer science, AI literacy, or career-connected technology instruction:

- CSF 3.1 - Occupations: Students research information technology occupations, work activities, tools and technology used, work environments, and knowledge and skills needed for success.
- CSF 13.1 - Social, Legal, and Ethical Issues: Students research social, legal, and ethical issues encountered by IT professionals and identify responsibilities when addressing technology problems.
- CSF 14.1 - Data Security: Students explain why data security is a priority and demonstrate understanding of confidentiality, availability, and integrity.
- CSF 14.2 - Security Breaches: Students demonstrate understanding of security breaches, enterprise-level security, encryption, protocols used to secure websites, and privacy considerations.
- CSF 14.3 - Security Practices: Students identify security practices for computer and network systems, including access control, encryption techniques, BIOS features, and malware response strategies.

### Use guidance

- Confirm final alignment against local district pacing, approved course placement, and teacher directions.
- Keep instruction defensive, ethical, age-appropriate, and classroom-safe.